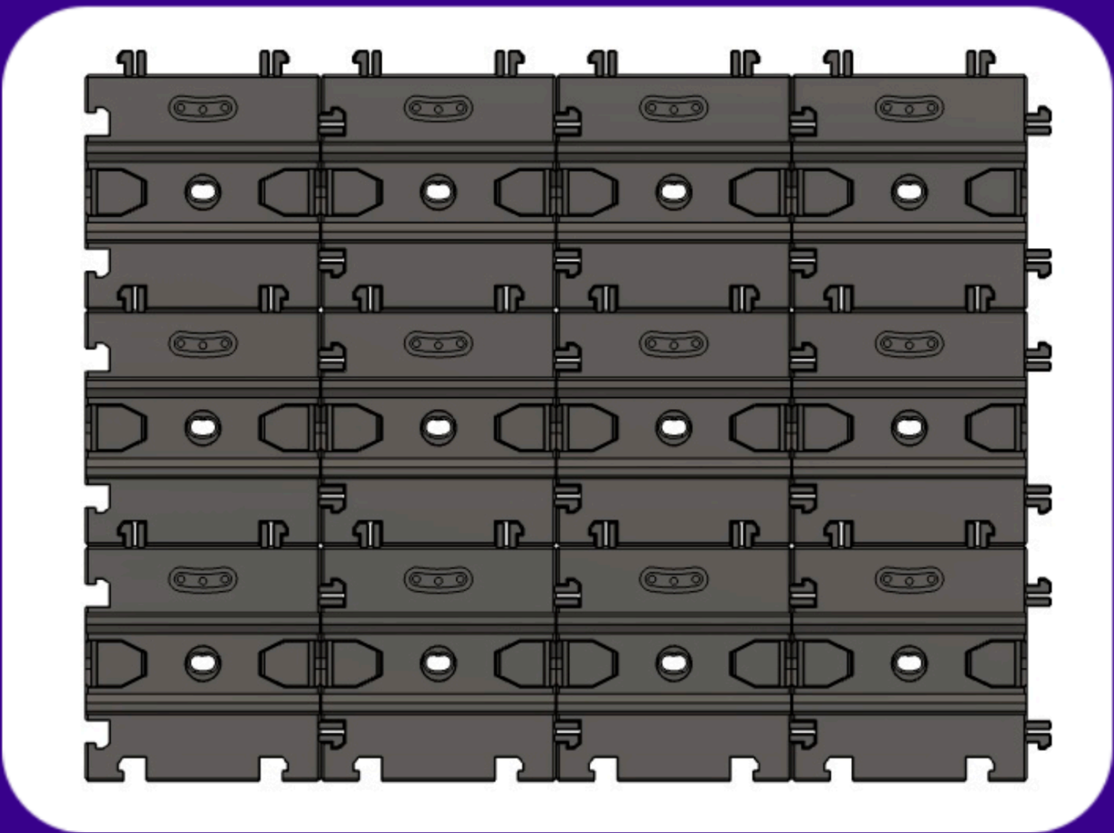
CARGO CONNECT: RAILS



User Guide

Resources, Print Settings and Assembly

Before You Start..

What's Included

The below files and documents are included within your download:

1. **Ready to Print (.3mf file) -** All the parts in one place and ready to print on Bambu Lab printers.
2. **User Guide (.pdf file) -** A complete 'User Guide' to help you print and assemble the model.
3. **Individual parts (.stl files) -** All the 3D printable parts you need to make the complete base model, compatible with most slicers.
4. **Assembly View Only file (Do Not Print) -** This file is to view the fully assembled model only and not for printing.
5. **Optional Parts (.zip folder) -** A selection of optional parts that can be used with the base model. Also includes an 'Archive' folder for any previous versions of the Cargo Connect: Rails *(see more information on the next page).*

What We Used

Printers:

Bambu Lab P1S

You can get the printers we used here (affiliate link)* **- Bambu Lab P1S**

Filaments:

Tiles - Bambu Lab PLA Basic Dark Grey

You can get the filaments we used here *(affiliate link*)* **- Bambu Lab Filament**

*Some links in this document are affiliate links, which means we may earn commission if you purchase through them.

Current Compatibility

Latest Updates:

The Cargo Connect: Rails and Cargo Containers have been updated as of the 28th February 2025. Any releases of the Cargo System before this date **will not** be compatible with the latest versions.

Already Printed a Previous Version?

Don't worry.. If you want to continue expanding your current system, you can use the Archive. Check out the information below!

Archive - Previous Versions

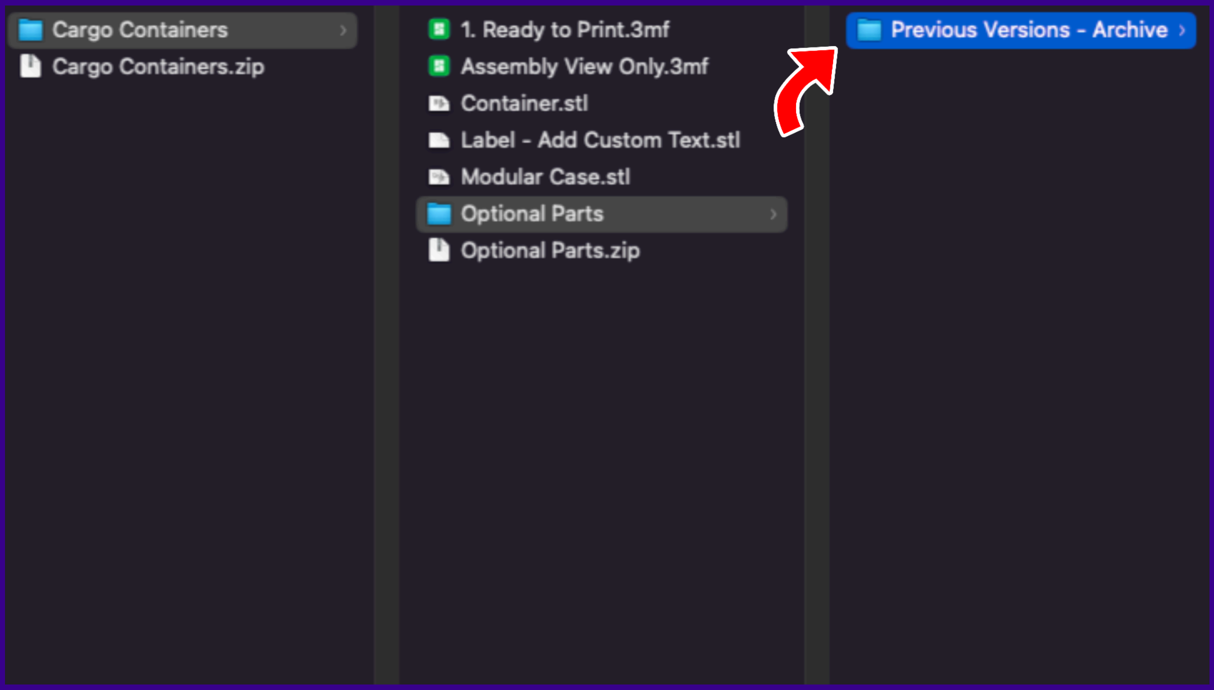
**Only relevant if you have already printed previous versions of the Cargo System*

The Cargo system is always growing and evolving. We've put a lot of work into future-proofing the system so new releases should play nicely with anything you've already printed.

However, from time to time, design upgrades may be necessary to improve the functionality of current and future components. While we strive to minimize disruption, these upgrades might mean that older printed parts are not compatible with the newest designs. *(See the 'Current Compatibility' section above for more information)*

To ensure you have options, if a component has previous versions that are not compatible with the latest updates, the downloadable files will include a **"Previous Versions - Archive"** zip folder *(see below)*.

This archive will contain all previous versions of that component, allowing you to either print the new upgraded versions or continue expanding your system with the version you've already started. *(Please be aware that while we will maintain the archive, older versions may not be compatible with the latest upgrades)*.



Print Settings

Important Note:
The information detailed in this 'Print Settings' document is the settings we found gave the best results for us. Your printer, materials and printing conditions are unique to you, and due to these widely varying aspects you will know what settings work best for your printers.

Ready to Print Project

A .3mf file named **'1.Ready to Print'** is included in the download. This file is a pre-prepared Bambu Studio project, which includes the settings we used.

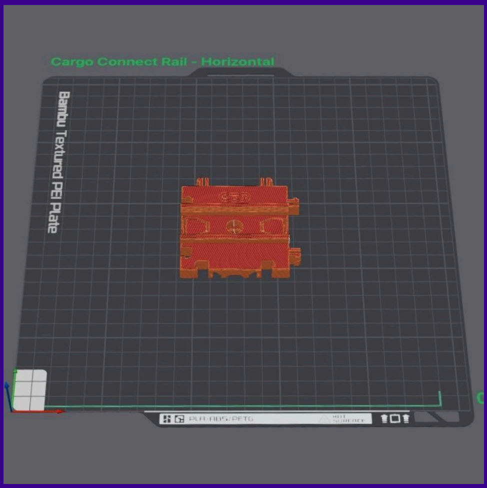
(As mentioned above, these settings worked best for us and for our needs, however, you may need to adjust these settings to suit your printing setup and needs)

Individual Part Settings

Horizontal Tile

Basic Settings:

- **Printer** - Bambu Lab P1S
- **Outer Walls** - 2
- **Infill** - 15%
- **Supports** - None
- **Brim** - Auto



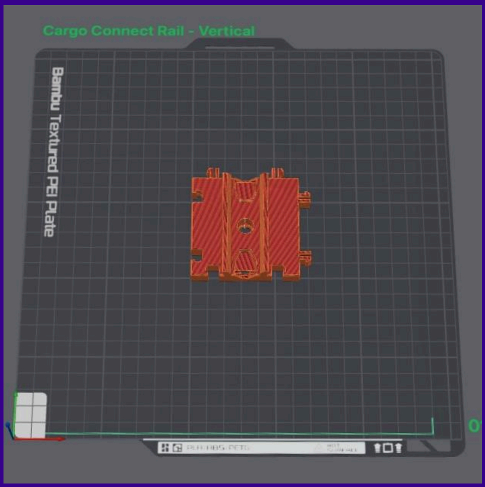
Estimations:

- **Print Time** - 41m 55s
- **Filament Used** - 21.82g

Vertical Tile

Basic Settings:

- **Printer** - Bambu Lab P1S
- **Outer Walls** - 2
- **Infill** - 15%
- **Supports** - None
- **Brim** - Auto



Estimations:

- **Print Time** - 41m 29s
- **Filament Used** - 21.76g

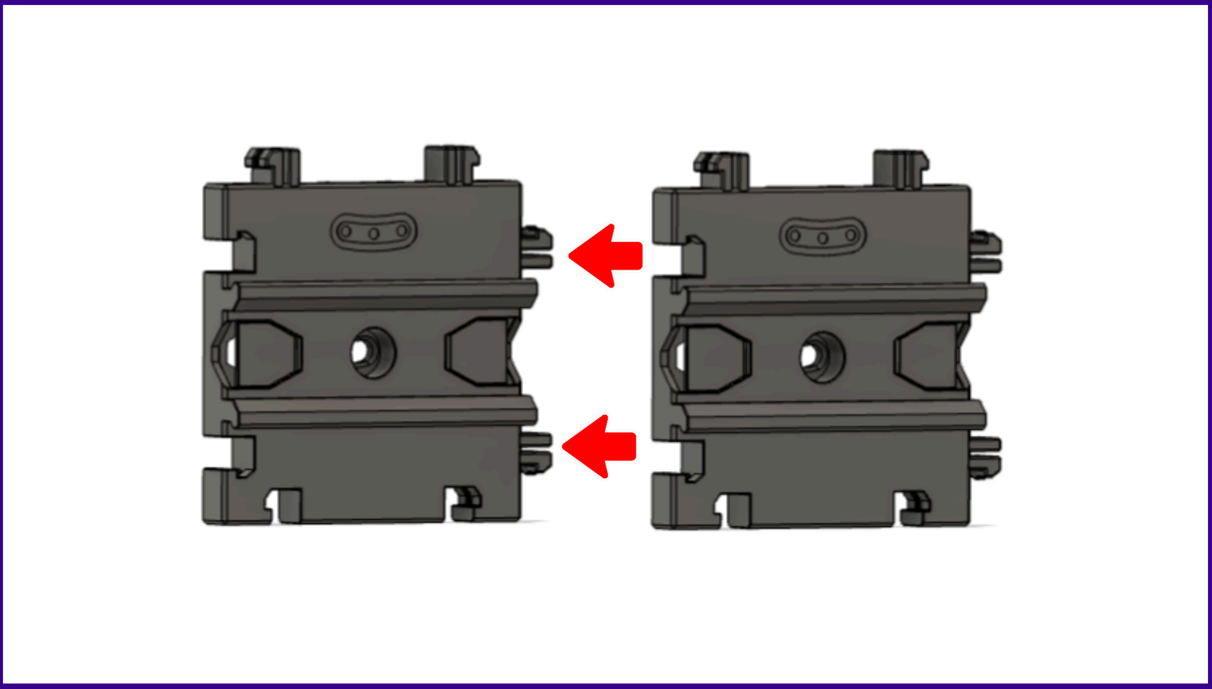
Assembly Guide

Important Note:

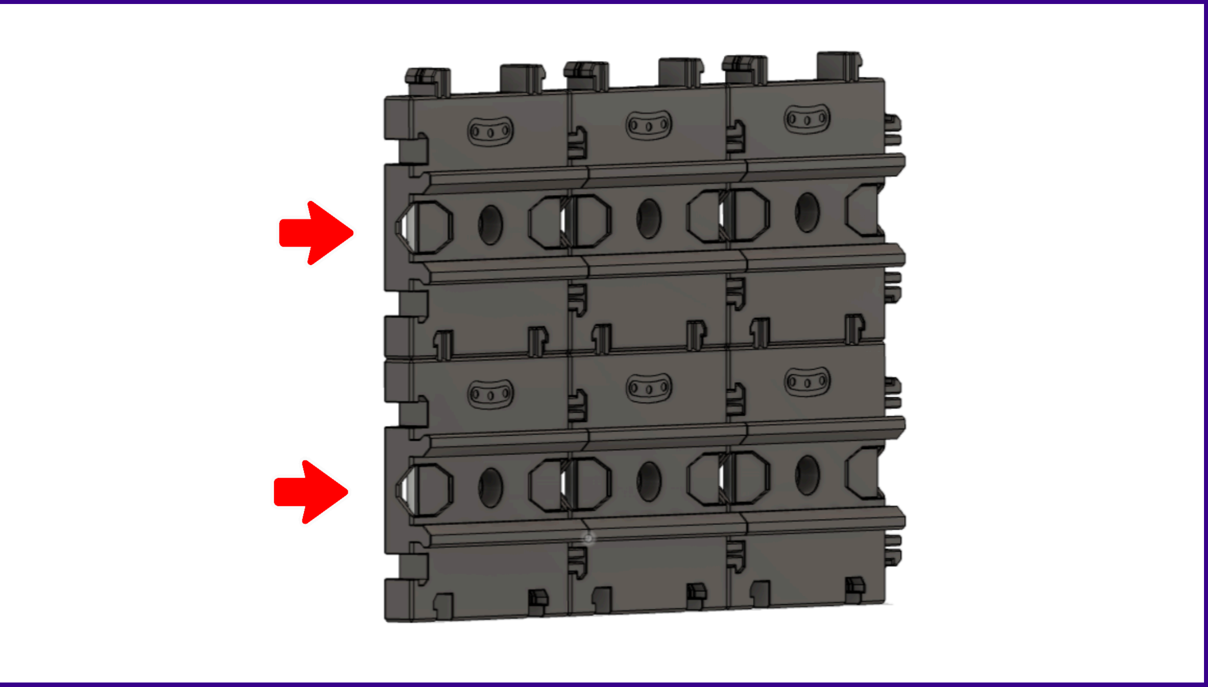
Once all parts are printed, remove any supports and check for any imperfections or poorly printed areas - *these may need post processing work or reprinting.*

Step 1

Each tile connects to another tile on any of the four sides using a 'Jigsaw' like connection – *(The tiles should be orientated so the rails line up together - this will allow any Cargo Connect compatible attachments to slide along them).*

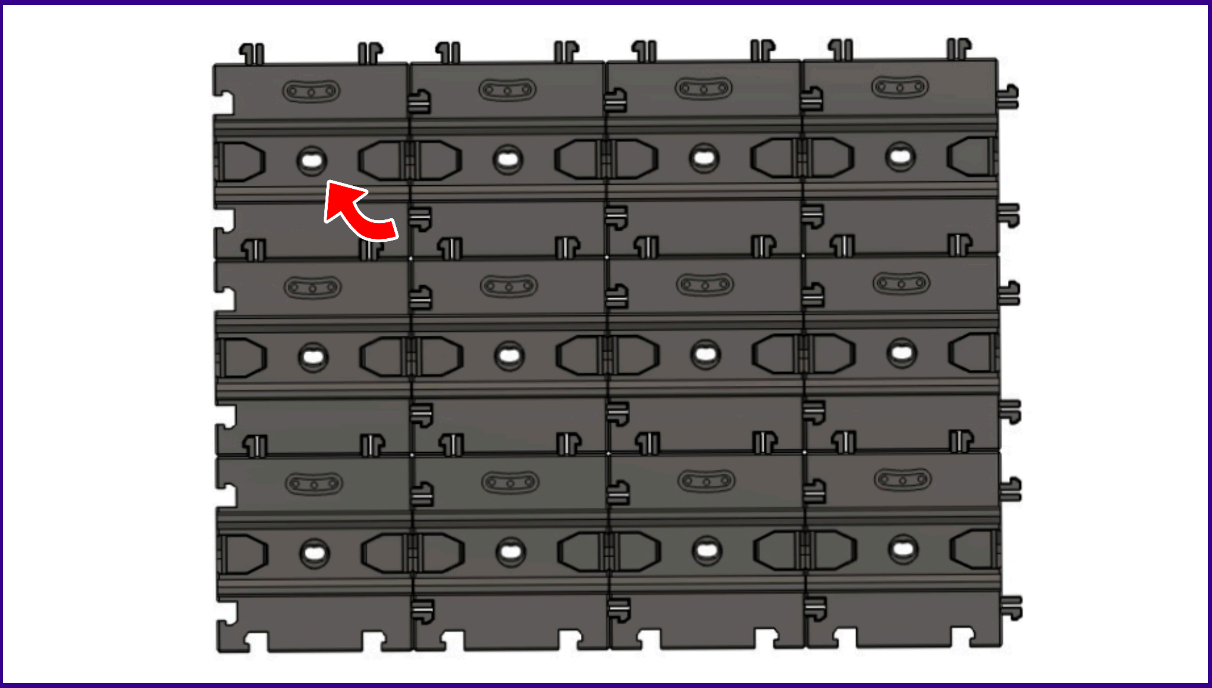


Lay the tiles on a hard flat surface and push them together until you hear the parts click together. Once properly connected, the tiles will be flush with each other and any compatible components will slide freely along them.



Step 2

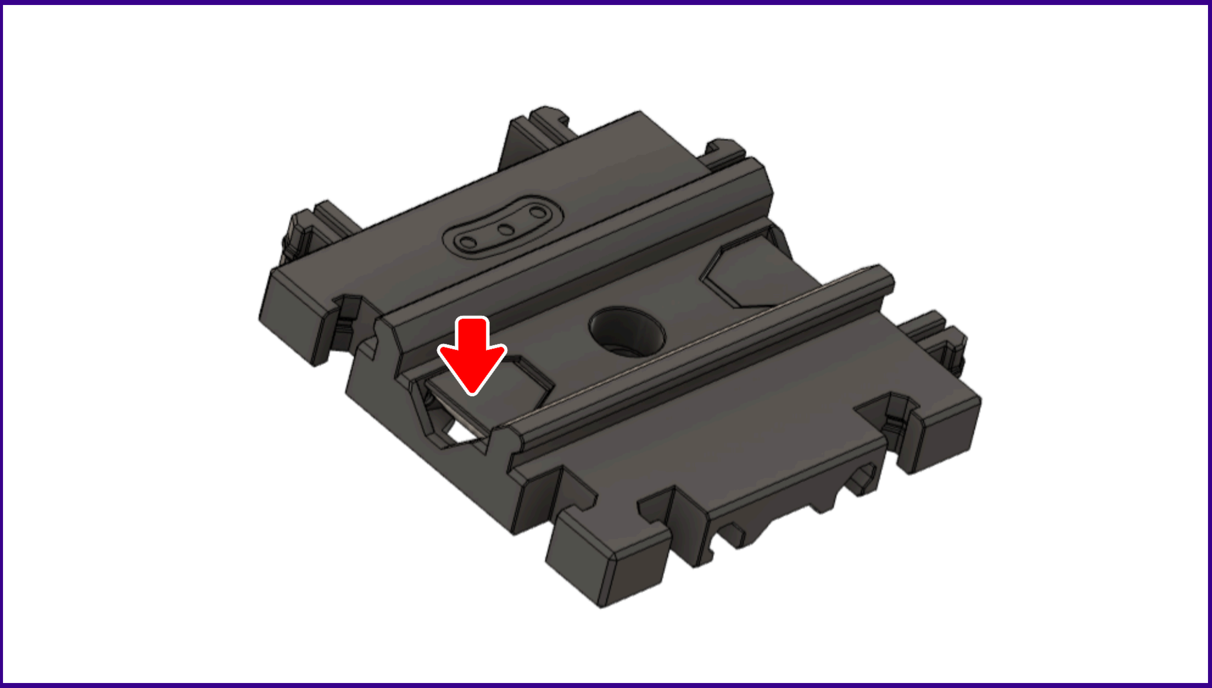
Each tile includes a screw hole for wall mounting. While the interlocking design provides inherent stability, **the number and placement of screws used is entirely at your discretion, subject to the strength of your wall and the intended weight of your stored items.** We recommend adding screws strategically until you are confident in the overall security of the installation.



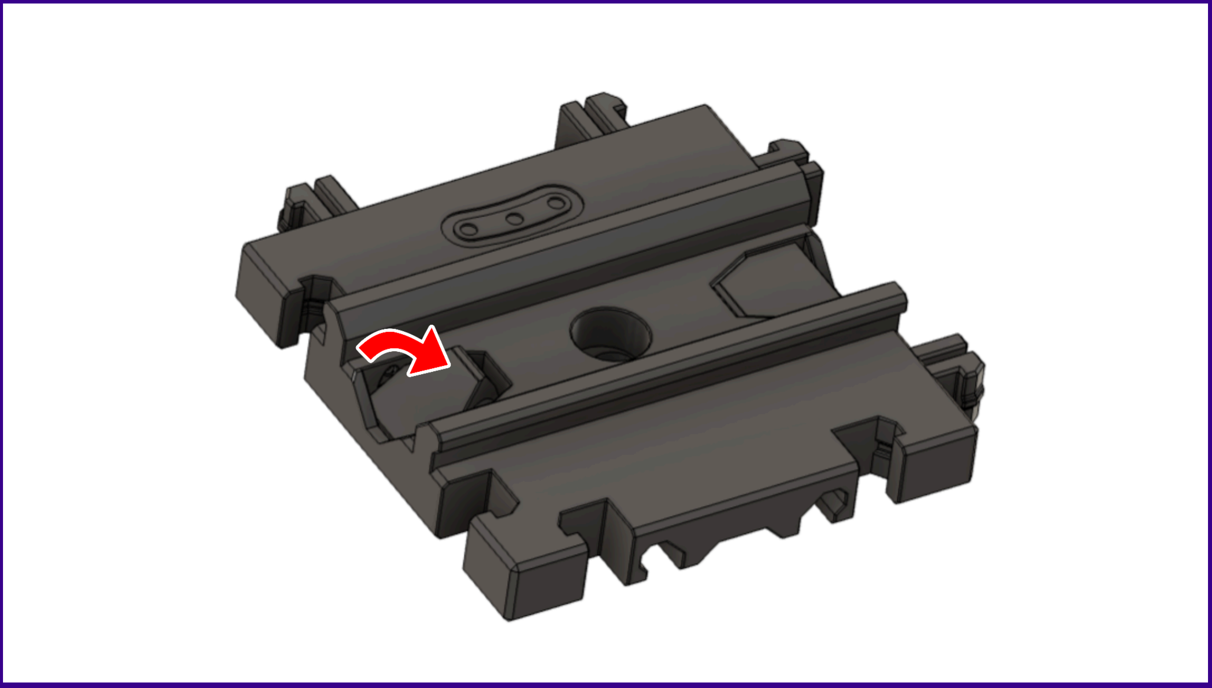
Step 3

Each tile has built in levers that act as 'stoppers', which will allow you to close off the edges or specific sections and stop any attached components sliding beyond that point.

To engage the built in 'stoppers' on the tile, push down into the recessed edge of the lever until you hear a click and the lever is now pointing upwards.



To disengage the stopper, simply push the top part back until you hear a click and it is flush with the top of the tile.



Important Note:

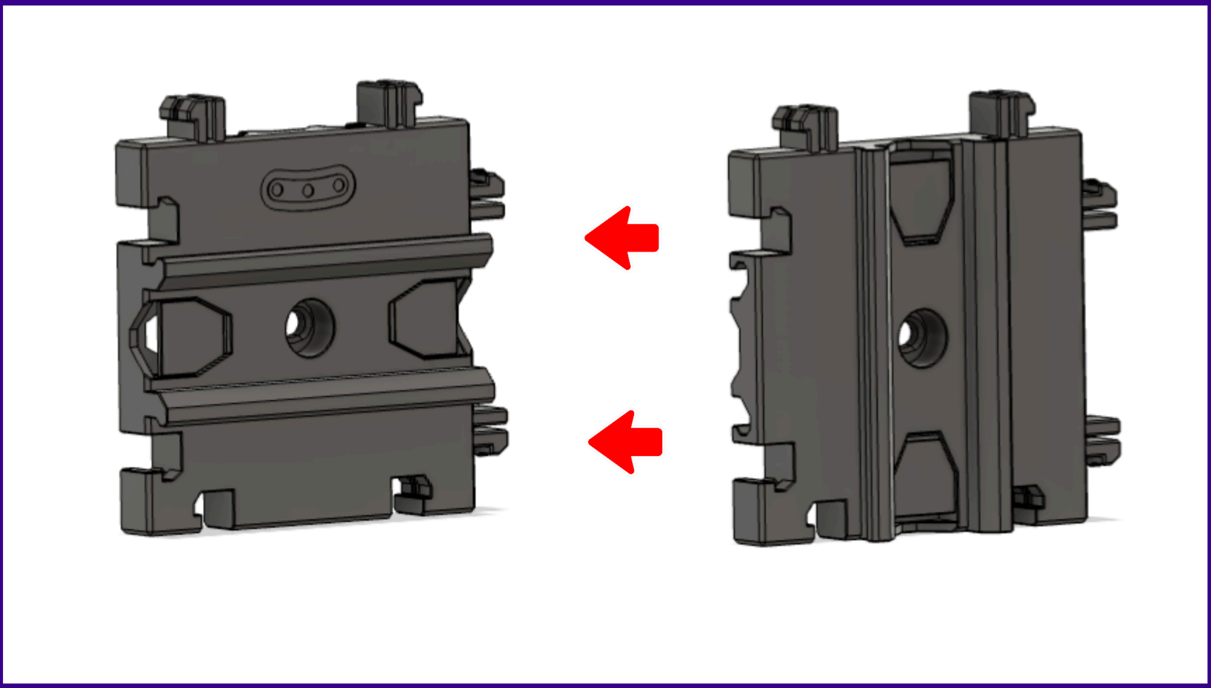
The levers may be slightly stiff on first use. If this is the case, engaging and disengaging the levers a few times will help 'bed them in' and make them easier to operate.

Vertical and Horizontal Rails

The Cargo Connect: Rails has two types of tile, Horizontal Rails or Vertical Rails. These rails are fully compatible with each other, which means they can interlock together using the connectors around each side, or they can slide on top of each other to create a multidirectional system.

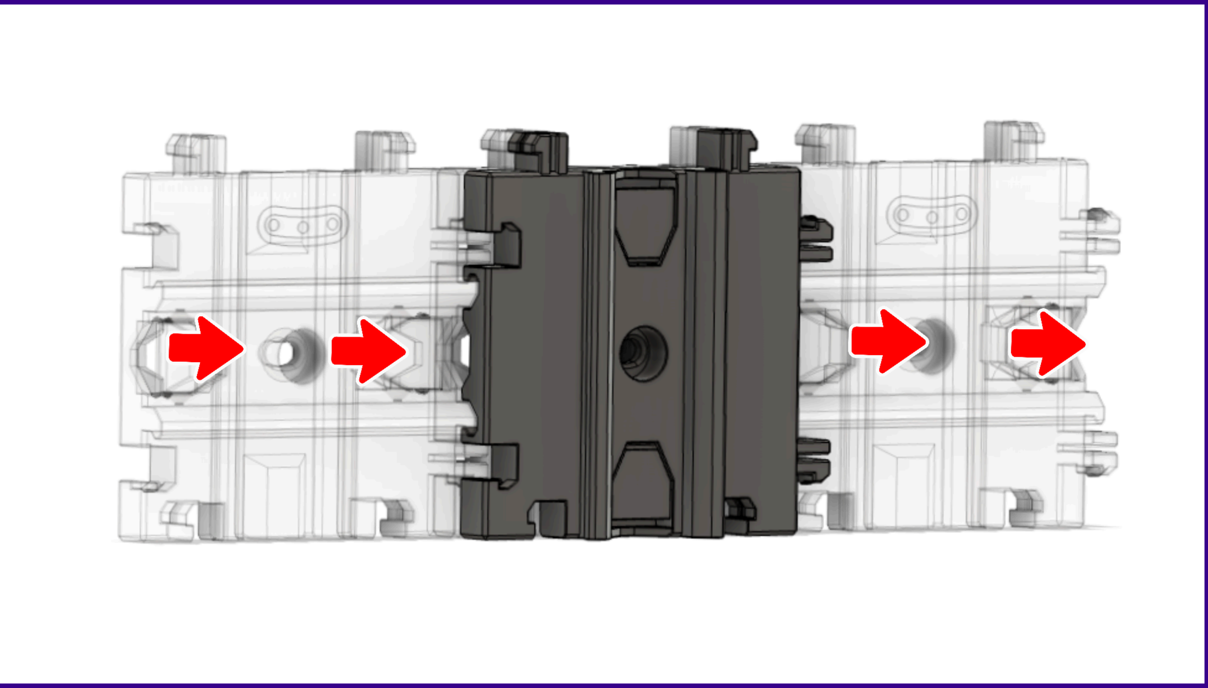
Interlocking Connections

As with Step 1, the Horizontal and Vertical tiles can simply push together using the connections around each side.



Sliding Rails

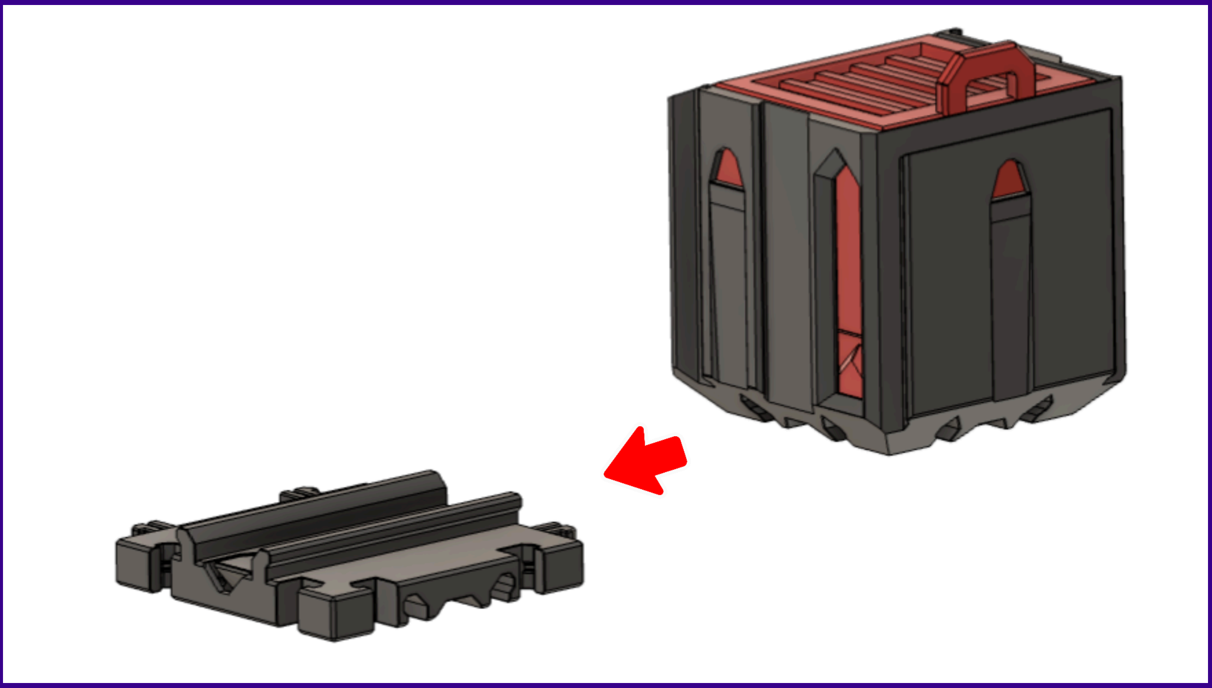
The Vertical Rails can slide onto the Horizontal Tiles and vice versa using the slots along the back of each tile. Simply slide a tile along the rails of the other as shown below.

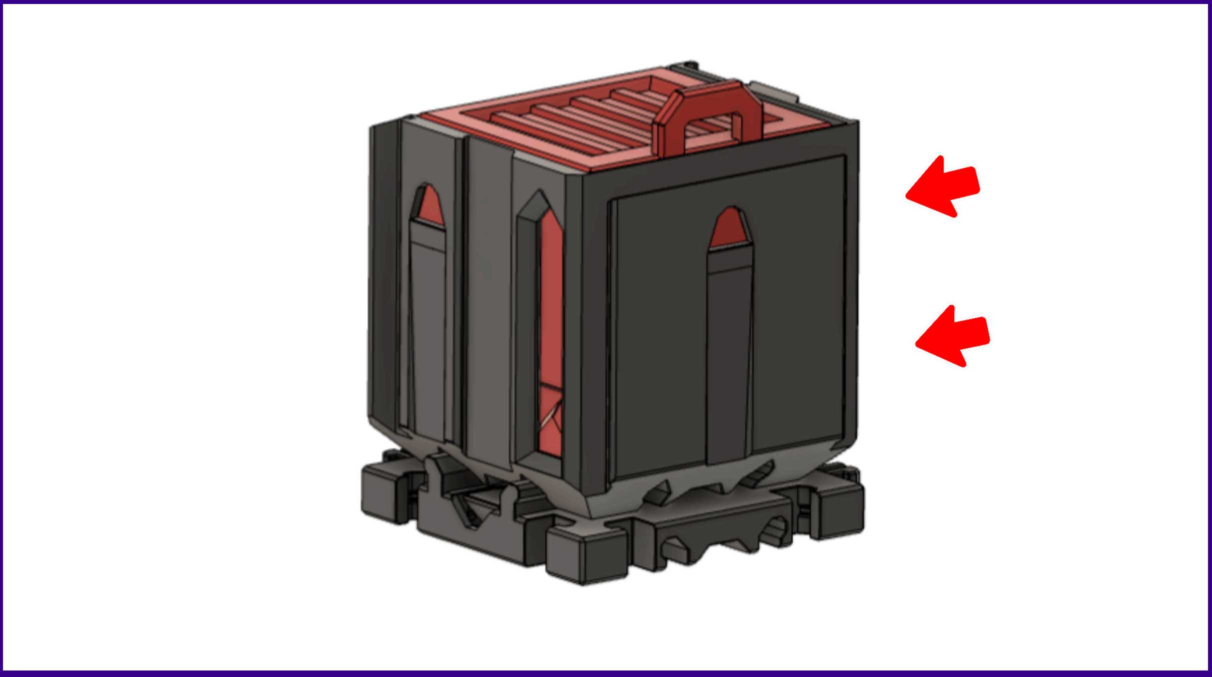


Compatible Components

The Cargo Connect: Rails allow compatible components to be attached to them. Once you have mounted your desired number of tiles, compatible components simply slide onto the rails as shown below.

You can find more compatible components on our Thangs page: **Play Conveyor - Thangs**

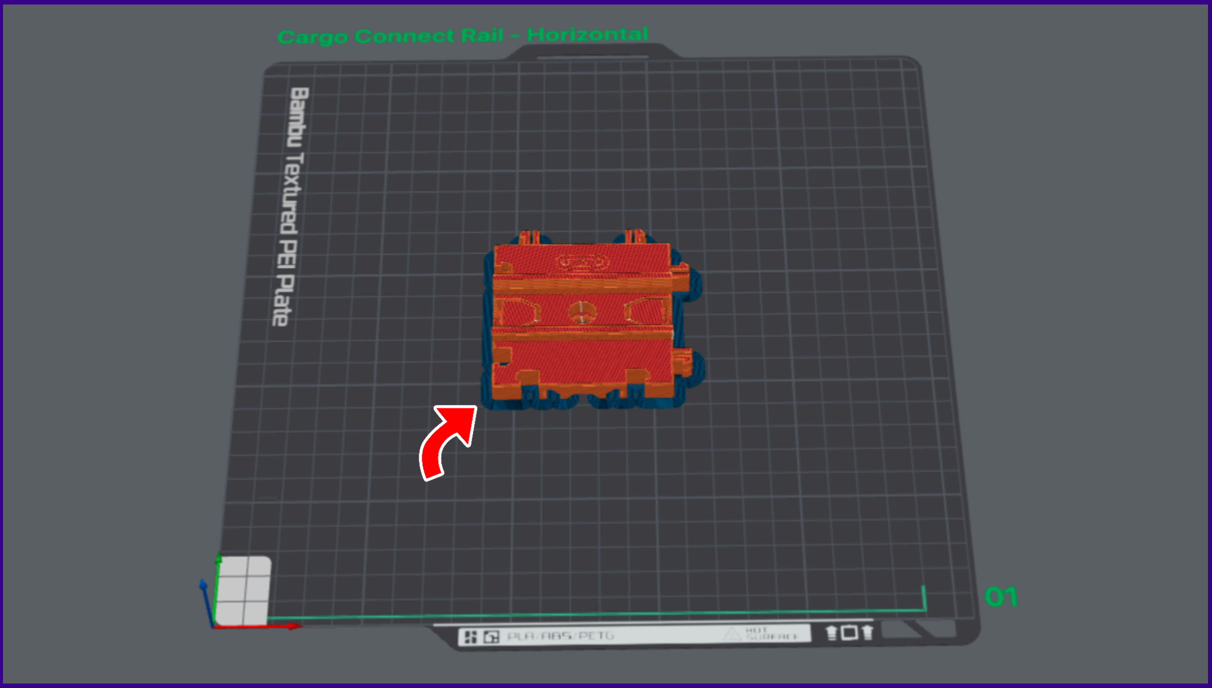




Troubleshooting

We put a lot of time into making our designs as easy to print and use as possible, testing extensively with various filaments and printers. That said, we understand that occasional issues can occur - below are some potential solutions that may be useful:

- If you are struggling with warping or bed adhesion when printing the tiles, adding a brim or 'brim ears' may help to solve this.



- Depending on your print quality and filament used, you may feel some friction when sliding the components along the rails. Firstly, check for any print imperfections or lumps of plastic that need removing. If the friction persists, we found adding a small amount of lubricant to the rails then sliding an attachment back and forth a few times helps this drastically *(Be sure to use lubricant suitable for your chosen filament).*